**CSA1006-SOFTWARE ENGINEERING**

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# CASE STUDY

## AI-Based Smart Airport Management System

## 1. Understanding the Problem Statement

### Problem Explanation

Modern international and domestic airports serve tens of thousands of passengers, flights, and luggage units every day. Passengers experience long waiting times at check-in counters and security zones, manual appointment and flight scheduling, misplaced or delayed baggage, unexpected flight delays, and inefficient communication between airport departments. The airport authority wants to develop an AI-Based Smart Airport Management System that enables automated flight scheduling, online and kiosk-based passenger check-in, real-time RFID baggage tracking, gate and boarding management, and AI-driven flight delay predictions.

### Objectives

- Develop an online and kiosk-based passenger check-in system.
- Automate and optimize flight scheduling using AI.
- Track passenger baggage in real time across all airport touchpoints.
- Manage gate allocation and boarding processes digitally.
- Reduce passenger waiting time and airport terminal congestion.
- Predict flight delays using AI based on weather and traffic data.
- Generate comprehensive airport performance and delay reports.

### Scope

- Passenger registration and identity verification
- Flight schedule and slot management
- Online and kiosk check-in system
- AI flight delay prediction engine
- RFID-based baggage tracking system
- Gate allocation and digital boarding management
- Real-time flight status notification system
- Centralized airport admin dashboard
- Operational report generation

## 2. Stakeholder Analysis

| Stakeholder | Responsibilities |
| --- | --- |
| Passengers | Register, check-in online/kiosk, track baggage, view flight status, obtain e-boarding pass. |
| Airline Staff / Ground Crew | Manage flight schedules, assist passenger check-in, verify boarding, oversee baggage loading. |
| Airport Security Staff | Verify passenger identity, manage security checkpoint queues, inspect flagged luggage. |
| Baggage Handlers | Scan and load luggage onto aircraft, update baggage transfer status across belts. |
| Air Traffic Controllers (ATC) | Coordinate runway usage, flight arrivals, departures, and emergency slot reallocations. |
| Airport Administrator | Manage system users, gate allocations, ground operations, and system security. |
| Airport Management | Monitor operational performance, flight delay analytics, and resource utilization. |

### Challenges

- Long waiting times at check-in counters and security gates
- Manual and static flight scheduling causing runway congestion
- Misplaced, delayed, or lost passenger baggage
- Poor communication during unexpected flight delays
- Delayed report generation for operational delays
- Difficulty tracking real-time passenger flow and gate allocation

## 3. Current System Analysis

### Existing Process

- Passengers visit airport check-in counters physically.
- Staff check-in passengers and issue paper boarding passes manually.
- Baggage is tagged with printed barcodes and scanned at limited manual checkpoints.
- Flight schedules are updated manually based on static timetables.
- Boarding gates are managed manually using handheld barcode scanners.

### Strengths

- Easy to understand by existing airport personnel
- No continuous internet or advanced cloud server dependency
- Low initial technology setup cost

### Limitations

- Time-consuming passenger processing
- Susceptible to human errors in baggage and check-in records
- Heavy reliance on paper tags and passes
- Duplicate or misplaced baggage files
- Long passenger waiting times during peak hours
- Difficult and slow performance report generation

### Problems

- Flight scheduling conflicts and runway delays
- Lost or misplaced passenger baggage
- Delayed departures due to slow manual boarding
- Poor resource and gate allocation
- Inefficient communication between ATC, staff, and passengers

## 4. Proposed Software Solution

### Proposed System

Develop a Web & Mobile-Based AI Smart Airport Management System that automates flight scheduling, passenger check-in, real-time RFID baggage tracking, boarding management, AI-based flight delay prediction, notifications, and automated report generation.

### Functional Modules

**Module 1: User & Passenger Management**

- Passenger Registration & Identity Verification
- Airline Ground Staff Login
- Security Staff Login
- Admin Login & Access Control

**Module 2: Flight Scheduling Module**

- Automated Flight Timetable Scheduling
- Runway & Gate Slot Allocation
- Real-Time Schedule Updates
- Airline Slot Coordination

**Module 3: Passenger Check-in Module**

- Online Web & Mobile Check-in
- Self-Service Kiosk Check-in Integration
- Digital Boarding Pass Generation (QR Code)
- Seat Selection & Baggage Allowance Check

**Module 4: RFID Baggage Tracking System**

- RFID Tag Assignment at Check-in
- Real-Time Luggage Tracking at Sorting & Loading
- Misplaced Baggage Automated Alert
- Passenger Mobile Baggage Tracking View

**Module 5: Gate & Boarding Management**

- Automated Gate Allocation
- E-Gate QR Code & Biometric Boarding Verification
- Priority Boarding Sequence Management
- Real-Time Passenger Headcount Reconciliation

**Module 6: AI Delay Prediction Module**

- Weather & Air Traffic Data Analysis
- Machine Learning Delay Prediction Model
- Automated Schedule Adjustment Suggestions
- Cascade Delay Warning System

**Module 7: Notification System**

- SMS & WhatsApp Flight Reminders
- Email Notifications & E-Boarding Pass Dispatch
- Real-Time Gate Change & Delay Alerts

**Module 8: Report Generation & Dashboard**

- Daily & Monthly Operational Reports
- Baggage Handling Performance Analytics
- Flight On-Time Performance Reports
- Passenger Traffic Statistics

## Functional Requirements

- User Registration & Verification
- Secure Multi-Factor Login
- Online & Kiosk Passenger Check-in
- Real-Time RFID Baggage Tracking
- AI Flight Delay Prediction
- Automated Gate & Flight Scheduling
- Boarding Pass Scanning & Gate Verification

• Real-Time Notifications
• Analytics & Report Generation

### Non-Functional Requirements

• Fast response time (Sub-second check-in & scanning)
• High availability (99.99% uptime for round-the-clock operations)
• Secure authentication & data encryption
• Reliable fault-tolerant database
• User-friendly interface for passengers and airport staff
• Scalability to handle peak holiday traffic volumes
• Real-time data backup and disaster recovery
• Privacy protection (GDPR compliance for passenger data)

### Software Engineering Principles Used

**Modularity**

Separate independent modules for flight scheduling, check-in, baggage tracking, delay prediction, and notifications.

**Scalability**

Supports increasing passenger numbers, airlines, and expanding terminal infrastructure.

**Maintainability**

Modules can be updated or serviced independently without bringing down the system.

**Reliability**

Automatic database backup, failover servers, and data validation.

**Usability**

Simple, multi-lingual interface for passengers and airport staff.

**Security**

Role-based access control, password encryption, and secure passenger data management.

## 5. SDLC Justification

### Selected Model

**Agile SDLC**

### Why Agile?

• Frequent requirement changes from civil aviation authorities and airlines
• Continuous feedback from airport staff, ATC, and ground crew
• Easy feature addition (e.g., integrating biometric gates or AI models)
• Faster software delivery through iterative sprint releases

- Continuous testing to ensure 100% operational safety and reliability

### Agile Process

Requirement Gathering
↓
Planning
↓
System Design
↓
Development
↓
Testing
↓
Deployment
↓
Maintenance

## 6. Expected Outcomes

### Benefits

- Faster check-in and boarding processing
- Reduced passenger waiting times and terminal queues
- Optimized flight scheduling and gate allocation
- Real-time visibility into baggage status
- Improved overall airport operational efficiency
- Better communication across all ground teams
- Increased passenger travel satisfaction

### Addressed Challenges

- Eliminates paper boarding passes and paper records
- Reduces flight scheduling and runway conflicts
- Provides real-time flight and baggage updates
- Improves multi-department communication
- Enhances operational report generation
- Enables better utilization of airport resources

### Risks

- Internet and network infrastructure dependency

• Initial software and hardware implementation cost
• User training required for airport staff and ground handlers
• Data privacy and security compliance concerns
• AI flight delay prediction errors during extreme weather events

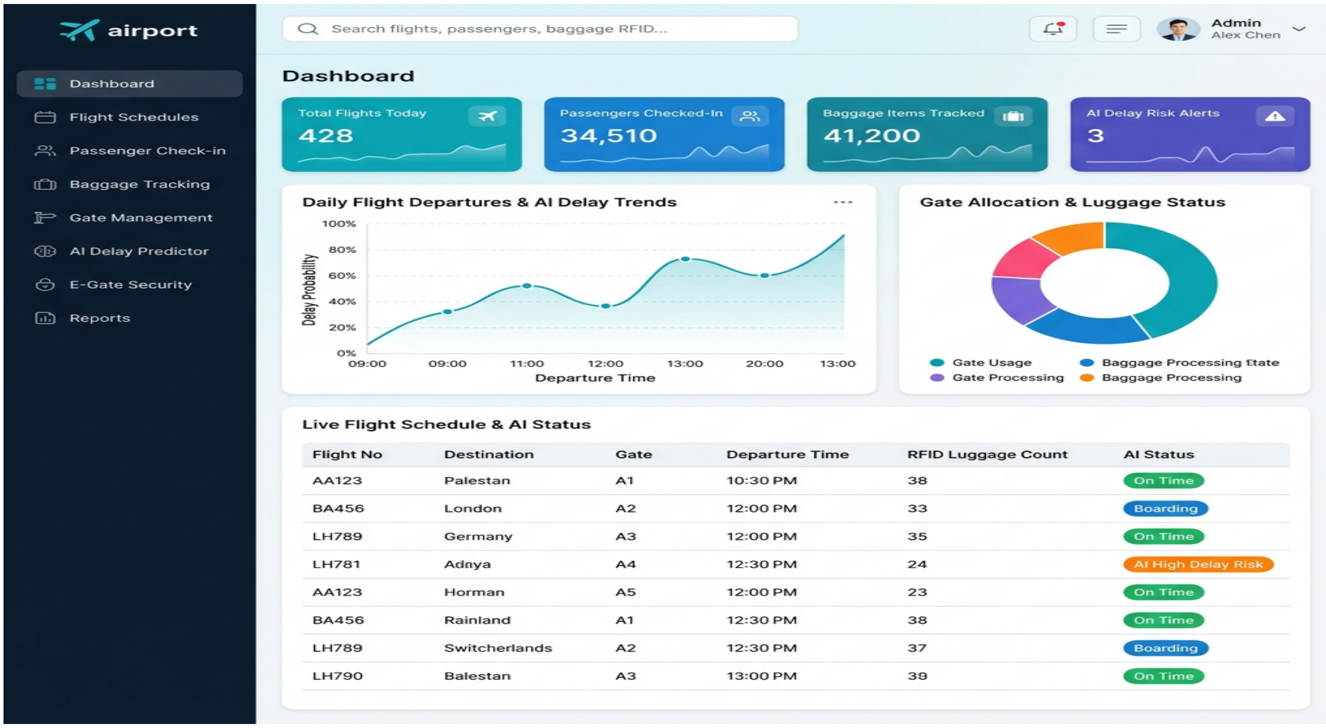
### Future Enhancements

• Mobile application with indoor airport navigation
• AI chatbot for passenger support and travel queries
• Biometric facial recognition for paperless e-gate boarding
• Autonomous baggage handling robot integration
• Predictive aircraft maintenance scheduling
• Automated luggage dimension and weight scanning kiosks

## Output Process Flow

Passenger
↓
Registers / Web Check-in
↓
Obtains Digital Boarding Pass & RFID Tag
↓
Baggage Tracked via RFID
↓
AI Predicts & Optimizes Schedule
↓
Passes Automated Boarding Gate
↓
Flight Departure
↓
Passenger Receives Notification
↓

## Reports Generated




## Conclusion

The AI-Based Smart Airport Management System provides an efficient, secure, and scalable solution for managing complex modern airport operations. By applying the Agile SDLC and core software engineering principles such as modularity, maintainability, scalability, usability, reliability, and security, the system optimizes flight scheduling, minimizes passenger check-in and waiting times, tracks baggage accurately in real-time, enhances communication, and predicts flight delays. Future enhancements including biometric facial recognition, AI chatbots, and autonomous robotics will make airport management even more intelligent, seamless, and efficient.